

# HTML End-to-End Lab Problems

Text • Formatting • Links • Images • Lists • Tables • Forms • Inputs

**Difficulty:** Advanced Practice

**Objective:** Build complete HTML pages by combining all concepts learned so far. Choose the correct element and attribute for each requirement instead of using tags mechanically.

## Concepts Covered

- Text and formatting: headings, paragraphs, br, hr, b, strong, i, em, mark, u, pre, code, blockquote, sub, sup and other taught formatting elements.
- Links and images: anchor/image elements, their taught attributes, and clickable images.
- Lists: ordered, unordered and description lists with their taught attributes.
- Tables: headings, rows, cells, captions, rowspan/colspan and other taught table attributes.
- Forms and inputs: form, label, input types, select/option, textarea, validation and all other attributes covered in class.

**Rule:** Do not use CSS or JavaScript. The goal is to demonstrate strong HTML structure, meaningful content, correct element selection, attributes, navigation, structured data and form design.

## Lab Problem 1: University Admission & Student Registration Portal

**Scenario:** A university wants a complete HTML-only admission portal where prospective students can understand the institution, explore programs, review fees, and submit an admission application.

### Requirements

1. Create a university landing section with a meaningful title, university name as h1, introductory paragraphs, campus image and useful links.
2. Create About University, Director's Message, Academic Programs, Campus Facilities, Student Life, Admissions and Contact sections.
3. Use an unordered list for program features and an ordered list for the admission process.
4. Use a description list for Eligibility, Merit List, Counselling, Scholarship and Admission Number.
5. Create a fee table with Program, Duration, Eligibility, Tuition Fee, Other Fee and Total Fee; include a caption and multiple rows.
6. Use rowspan/colspan where a genuine data relationship requires merged cells, if taught.
7. Create an Important Dates section and highlight the most important deadline using mark.
8. Create a complete admission form for personal details, contact details, academic information, selected program, category, gender, date of birth, address, document upload if taught, declaration and submission.
9. Use appropriate input types, labels, name/value attributes and validation such as required, minlength, maxlength, min, max, step, pattern, placeholder and autocomplete where appropriate.
10. Use select/option for predefined choices and radio buttons/checkboxes according to whether one or multiple choices are needed.
11. Use textarea for address or other long-form information.
12. Add contact links, at least two meaningful images and at least one clickable image.
13. Use blockquote for a message from the university administration.

**Challenge:** Integrate informational content, structured data, navigation and a substantial form into one coherent portal rather than creating disconnected tag demonstrations.

## Lab Problem 2: Healthcare Appointment & Patient Intake Portal

**Scenario:** A private clinic wants a patient-facing HTML portal where visitors can learn about the clinic, view doctors and timings, and request an appointment through a detailed intake form.

### Requirements

1. Create a clinic introduction with name, description, relevant image and useful links.
2. Create Doctors & Specialties using headings, paragraphs and a list of specialties.
3. Create a doctor table with Doctor Name, Specialty, Available Days, Consultation Type and Consultation Fee.
4. Use a caption and proper table headings; use merged cells only where logically justified.
5. Create a description list defining New Patient, Follow-up, Consultation, Emergency and Teleconsultation.
6. Create an ordered list showing the appointment process and an unordered list of information/documents a patient should keep ready.

7. Create a detailed form for patient name, date of birth, email, phone, gender, doctor, specialty, appointment date/time, visit type, communication preference, patient ID and reason for visit.
8. Use appropriate input types instead of using text for everything.
9. Use required and other validation attributes appropriately, including sensible date/number constraints where relevant.
10. Use radio buttons for single-choice decisions and checkboxes for independent selections/consents.
11. Use select/option for doctor, specialty and visit type where appropriate.
12. Use textarea for symptoms/reason or additional information.
13. Include consent and suitable submit/reset controls.
14. Add clinic contact/emergency links, an image link, and a short patient-care blockquote.

**Challenge:** Design the form for different kinds of data while keeping the surrounding information organized with the correct lists and tables.

### Lab Problem 3: Online Learning Platform — Course Catalog & Enrollment

**Scenario:** An education company wants a complete course portal. Students should explore courses, compare them, understand prerequisites, and enroll through a registration form.

#### Requirements

1. Create a platform introduction with title, h1, description, image and navigation links.
2. Create a catalog with at least six courses using headings and lists.
3. Use an unordered list for course features and an ordered list for the enrollment journey.
4. Use a description list for Beginner, Intermediate, Advanced, Prerequisite, Certification and Live Session.
5. Create a course comparison table with Course, Duration, Level, Mode, Mentor, Fee and Certificate; include at least six rows and a caption.
6. Create a featured-course section with detailed paragraphs, image, formatting, blockquote and a pre/code example relevant to programming.
7. Make the featured course image clickable.
8. Create an enrollment form for name, email, password, mobile, date of birth, course, batch, learning mode, technical interests and learning goals.
9. Use radio buttons, checkboxes, select/option and textarea correctly.
10. Use a URL input for GitHub/portfolio if taught and apply suitable validation.
11. Use meaningful name/value attributes and validation such as required, minlength, maxlength, min, max, step, pattern, placeholder and autocomplete where appropriate.
12. Include a terms/accuracy checkbox and Submit/Reset controls.
13. Add internal and external links and use strong/em/mark purposefully.

**Challenge:** Combine a catalog, comparison table, detailed course section and enrollment workflow into one consistent learning platform.

### Lab Problem 4: Travel Booking & Tour Package Explorer

**Scenario:** A travel company wants a complete HTML travel page where customers can explore destinations, compare tour packages, understand package terms and submit a booking request.

#### Requirements

1. Create a travel company introduction with main heading, description, company image and useful links.
2. Feature at least four destinations with images and descriptive paragraphs.
3. Make at least two destination images clickable and connect them to relevant information pages.
4. Create a Tour Packages table with Package, Destination, Duration, Hotel Type, Transport, Price Per Person and Availability; include at least six rows and a caption.
5. Use rowspan/colspan if a genuine package relationship requires it and the attribute was taught.
6. Create an ordered booking-process list.

7. Create an unordered list for items included in every package.
8. Create a description list defining Per Person, Twin Sharing, Non-Refundable, Transfer, Sightseeing and Package Upgrade.
9. Create a detailed travel tips section using headings, paragraphs and purposeful formatting.
10. Create a booking form for customer name, email, phone, destination, package, travel date, number of travelers, room preference, transport preference and special requests.
11. Use appropriate number/date inputs, constraints and select controls.
12. Use radio buttons for single-choice preferences and checkboxes for independent add-ons.
13. Use textarea for special requests and include confirmation plus Submit/Reset controls.
14. Add contact/support/package links and at least four meaningful images.

**Challenge:** Make the page function conceptually like a booking system while correctly combining lists, tables, images, links and forms.

## Lab Problem 5: FINAL LAB — E-Commerce Store Product, Cart & Checkout

**Scenario:** A new online store wants a complete HTML prototype of its shopping experience. A visitor should understand the store, explore products, review product comparisons and cart data, and provide checkout information.

### Requirements

1. Create a store introduction with meaningful title, h1, description, store/logo image and navigation links.
2. Create product categories using headings and unordered lists.
3. Create at least six featured products; each needs an image, name, short description and relevant link.
4. Make at least three product images clickable and connect each to a different product/detail destination.
5. Create a product comparison table with Product, Category, Key Feature, Storage/Capacity, Warranty, Price and Rating; include at least six products and a caption.
6. Create an ordered list describing the shopping process: browse, select, review, checkout and confirmation.
7. Create a description list explaining MRP, Discount, Coupon, Delivery Fee, Return Window and Warranty.
8. Create a Cart/Order Summary table with Product, Quantity, Unit Price, Discount and Total.
9. Create a complete checkout form containing customer information, delivery address, city/state/country, postal code, delivery method, payment method, coupon code, billing-address option and delivery instructions.
10. Use appropriate input types and constraints for email, phone, number, date and other data.
11. Use select/option for predefined choices; radio buttons for single-choice options; checkboxes for independent confirmations/add-ons.
12. Use textarea for delivery instructions.
13. Use required, minlength, maxlength, min, max, step, pattern, placeholder, autocomplete and other taught attributes where they genuinely improve the form.
14. Use meaningful name/value attributes throughout.
15. Include terms acceptance and order confirmation plus Place Order and Reset controls.
16. Add Continue Shopping and other useful links, product/store images, a customer blockquote and a pre/code product specification.
17. Use previously learned formatting and list/table concepts where appropriate.
18. Do not use CSS or JavaScript.

**Challenge:** This is the most comprehensive lab. Connect content structure, product media, navigation, lists, comparison/cart tables and a multi-section checkout form into one consistent HTML document.

## Final Quality Checklist

- Choose the correct HTML element for the type of information being represented.
- Use ordered lists for sequences, unordered lists for collections, and description lists for terms and definitions.
- Use tables for structured multi-record data, not for general page layout.
- Use table headers, captions and merged cells correctly where appropriate.
- Every form control has a meaningful label and appropriate input type.
- Use name/value attributes consistently and logically.
- Use validation attributes according to the actual data requirements.
- Radio buttons and checkboxes are used for the correct type of choice.
- Links and images use the attributes taught in class and meaningful destinations/alternative text.
- Use formatting elements purposefully.
- Keep HTML readable, well-indented and logically structured.
- Do not use CSS or JavaScript.

## Suggested Folder Structure

```
HTML/  
  End_to_End_Lab/  
    problem-01.html  
    problem-02.html  
    problem-03.html  
    problem-04.html  
    problem-05.html
```